SVM Compute Core — Full-Chip Design Summary (m5/v9: Final Harden)

Project: Multi-Class Cardiac Arrhythmia Detection — Caravel chipIgnite Tape-Out **Technology:** sky130A / sky130_fd_sc_hd **Flow:** OpenLane 2 v2.3.10 Classic **Architecture:** Batch v8/v9 — host pre-loads SV + input matrix; ASIC classifies autonomously **RTL freeze:** m5/rt1 v9 — NUM_SV=500, alpha_addr[8:0], reg_alpha_wr[24:0]

Component Summary

svm_compute_core (job 91966, v9)

Metric	Value
Clock	40 MHz (25 ns), TT corner clean
Setup WNS (TT)	+7.83 ns — 0 violations
Hold WNS (TT)	+0.30 ns — 0 violations
DRC	0 violations
Active power	~66 mW
Inference time	3.23 ms / beat (500 SVs × 256 dim at 40 MHz)
Avg power (80 bpm)	**0.284 mW** (0.431% duty cycle)
Die area	2500 × 2500 µm (~14% utilization)
GDS	226 MB
LEF	94 KB
GL netlist	13 MB

user_project_wrapper (job 91967)

Metric	Value
Die area	2920 × 3520 µm (Caravel fixed, FP_DEF_TEMPLATE)
Macro	u_svm at (253, 554) N — 2500 × 2500 µm footprint
Clock	wb_clk_i (Caravel), gated to svm_gclk via ICG
CTS	Disabled (RUN_CTS: 0)
DRC	11,923 Magic DRC (boundary artifacts — acceptable)
LVS	1,683 errors (boundary artifacts — acceptable)
GDS	230 MB
LEF	195 KB
GL netlist	78 KB

Functional Results

Implementation	Accuracy	SVs	Notes
sklearn OVR (float)	97.67%	416 total (unlimited)	float precision
ASIC binary OVR (Q6.10)	97.67%	500 total (100×5)	gamma=0.25, C=1.0

Zero accuracy gap — ASIC exactly matches sklearn on all 300 test samples. See Testing Set section for per-class breakdown.

Training Set

Parameter	Value
Source	MIT-BIH Arrhythmia Database (PhysioNet)
Split	80% train / 20% test — stratified, `random_state=42`
Beats per class	240
Classes	Normal (N), PVC, AFib, VT, SVT
Total training beats	**1,200**
Feature dimension	256 (128 single-beat + 64 10-beat mean + 64 RR history)
Precision	Q6.10 fixed-point (16-bit signed)

Testing Set

Parameter	Value
Source	MIT-BIH Arrhythmia Database (PhysioNet) — held-out 20%
Beats per class	60
Total test beats	**300**
sklearn accuracy	97.67% (293/300)
ASIC accuracy	97.67% (293/300) — zero gap vs. float

Per-class test results:

Class	Correct	Accuracy
Normal (N)	60/60	100.0%
PVC	60/60	100.0%
AFib	60/60	100.0%
VT	56/60	93.3%
SVT	57/60	95.0%

Batch Architecture (v8/v9)

Off-chip RAM Bus

Signal	Pin	Direction	Description
`ram_addr[18:0]`	GPIO[28:10]	ASIC out	{row[10:0], col[7:0]}
`ram_ren`	GPIO[29]	ASIC out	Read strobe
`ram_rdata[15:0]`	LA[15:0]	Host in→ASIC	Data valid after `RAM_LATENCY` cycles

Address layout: rows 0..499 = SV matrix; rows 500..1499 = input matrix.

`RAM_LATENCY` parameter — configures the number of clock cycles between `ram_ren` assertion and valid `ram_rdata`. Default is 1 (same-cycle combinational SRAM model used in cosim). Set to 3 for the IS61WV51216 async SRAM (10 ns access at 40 MHz → 3-cycle pipeline). The core inserts wait states automatically; the host does not need to pad responses. Verified by `svm_ram_latency_tb.sv` (10-beat, LAT=3, PASS in ~208 cycles/beat).

What the Host Does

```
MCU (low-power, continuous)
     │
     │  1. Collect 1000 heartbeats (250 Hz ECG → feature extraction)
     │  2. Load SV matrix  (500 SVs × 256 features) → SRAM rows 0..499
     │  3. Load input matrix (1000 beats × 256 features) → SRAM rows 500..1499
     │  4. Write NUM_SAMPLES = 1000, write NUM_SV_0–4 = 100 each
     │  5. Write alpha coefficients via ALPHA_WR (Wishbone 0x28)
     │  6. Fire CONTROL[start]
     │
     ▼  ASIC takes over (timing shown for RAM_LATENCY=1 / RAM_LATENCY=3):
     ├── LOAD_INPUT per beat:    256 / 768 cycles  (reads from SRAM rows 500+)
     ├── COMPUTE_DIST per SV:   258 / 770 cycles  (reads SV from SRAM rows 0–499)
     ├── COMPUTE_KERNEL: ~18 cycles (Horner LUT exp approximation, LAT-independent)
     └── WRITE_CLASS: argmax+alpha → sample_rdy (IRQ[0]) per beat
                  last beat → done (IRQ[1])
```

Inference time scales linearly with `RAM_LATENCY`. At 40 MHz: - LAT=1 (cosim default): **3.23 ms / beat** (500 SVs × 256 dim) - LAT=3 (IS61WV51216 SRAM): ~**9.7 ms / beat** — still well within the 750 ms heartbeat period

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Wishbone Register Map (base `0x3000_0000`)

Offset	Name	R/W	Description
+0x04	CONTROL	RW	[0]=start [1]=vbatt_ok [2]=vbatt_warn
+0x08	STATUS	RO	[0]=done [1]=error [5:2]=error_code [8:6]=class [9]=sample_rdy
+0x0C	NUM_SAMPLES	RW	[9:0] beats in batch (1–1000)
+0x10–+0x20	NUM_SV[0–4]	RW	[7:0] SVs per class (max 100 each)
+0x24	PARAM_WR	WO	[19]=en [18:16]=addr [15:0]=data (, C, bias)
+0x28	ALPHA_WR	WO	[24:16]=sv_global_idx (9-bit) [15:0]=alpha Q6.10

Full-Chip Power Estimate

Subsystem	Active Power	Duty Cycle	Avg Power
svm_compute_core (batch)	66 mW	0.431% (3.23 ms / 750 ms)	**0.284 mW**
Caravel management SoC	~5 mW	~5%	~0.25 mW
ECG frontend (analog)	~0.5 mW	100%	0.5 mW
BLE (optional, logging)	~10 mW	~0.1%	~0.01 mW
Total estimated	—	—	~**1.04 mW**

Battery budget: 200 mAh @ 3.7V = 740 mWh → **740 mWh / 1.04 mW 711 hours (~29.6 days)**. 14-day target met with 2.1× margin. SVM core alone: ~2606 hours (~108 days).

Caravel Submission Artifacts

Repository: github.com/adamleehandwerger/caravel_svm_project
GDS Release: v2.0-hardened (226 MB core / 230 MB wrapper)

Layout artifacts:

File	Size	Job	Status
`gds/svm_compute_core.gds`	226 MB	91966	
`gds/user_project_wrapper.gds`	230 MB	91967	
`lef/svm_compute_core.lef`	94 KB	91966	
`lef/user_project_wrapper.lef`	195 KB	91967	

Verilog artifacts:

File	Size	Job	Status
`verilog/gl/svm_compute_core.v`	13 MB	91966	
`verilog/gl/user_project_wrapper.v`	78 KB	91967	
`verilog/rtl/svm_compute_core.sv`	—	v9	
`verilog/rtl/user_project_wrapper.sv`	—	v9	

Acknowledgments

Place-and-route was performed on **Orca**, Portland State University's high-performance computing cluster, using SLURM batch jobs with OpenLane 2 v2.3.10 inside a Singularity container. We thank the PSU Research Computing team for providing access to the GPU and CPU nodes that made the multi-hour OpenLane runs feasible.

Feature Extraction References

The 256-dim multi-scale feature vector follows established AAMI EC57 beat classification conventions:

Feature group	Dims	Reference
Single-beat morphology (±64 samples, amplitude-norm)	128	de Chazal P, O'Dwyer M, Reilly RB. "Automatic classification of heartbeats using ECG morphology and heartbeat interval features." *IEEE Trans Biomed Eng* 51(7):1196-206, 2004. DOI: 10.1109/TBME.2004.827359
10-beat mean morphology template	64	de Chazal P, Reilly RB. "A patient-adapting heartbeat classifier using ECG morphology and heartbeat interval features." *IEEE Trans Biomed Eng* 53(12):2535-43, 2006. DOI: 10.1109/TBME.2006.883802

Feature group	Dims	Reference
RR-interval history (99 intervals → 64 pts, norm to NORMAL_RR=308 ms)	64	Llamedo M, Martínez JP. “Heartbeat classification using feature selection driven by database generalization criteria.” *IEEE Trans Biomed Eng* 58(3):616-25, 2011. DOI: 10.1109/TBME.2010.2068048

Standard: AAMI ANSI EC57:2012 — Performance Requirements for Ambulatory ECG Analysers.
Dataset: PhysioNet MIT-BIH Arrhythmia Database (Moody GB, Mark RG, 2001).
DOI: 10.13026/C2F305

Document version: m5/v9 · 2026-05-28 — RAM_LATENCY parameter added; cosim 97.67% = sklearn; Appendix A added; Appendix B added

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Appendix A — Runtime Model Reload Procedure

The ASIC is fully runtime-reprogrammable. A new trained SVM model can be loaded without resetting the chip or re-synthesizing. The on-chip alpha table and the off-chip SRAM are simply overwritten in place.

What constitutes a “model”

Parameter	Location	Size
SV matrix (500 × 256 × 16-bit features)	Off-chip SRAM, rows 0–499	256 KB
Alpha coefficients (500 × 16-bit)	On-chip `alpha_table[]` registers	8 KB
Gamma (, Q6.10)	On-chip `gamma_reg` shadow register	2 bytes
SV counts per class (`NUM_SV[0–4]`)	On-chip Wishbone registers	5 bytes

Reload sequence

Perform these steps while the ASIC is idle (`STATUS[done]=1` or after reset). Do **not** fire `CONTROL[start]` until all steps are complete.

Step 1 — Write new SV matrix to off-chip SRAM

Write 500 rows × 256 columns of Q6.10 feature values to SRAM address rows 0–499. Address encoding: `addr[18:0] = {sv_global_idx[10:0], feat_dim[7:0]}`. This is a bulk SRAM write from the MCU and does not involve the Wishbone bus.

```
for sv in range(500):
    for feat in range(256):
        sram_write(addr=(sv << 8) | feat, data=sv_matrix[sv][feat])
```

Step 2 — Write new alpha coefficients via ALPHA_WR

Write all 500 alpha values over Wishbone. Each write encodes the SV global index and the alpha value in a single 32-bit word:

```
WB base = 0x3000_0000
ALPHA_WR offset = 0x28

for idx in range(500):
    word = (idx << 16) | (alpha[idx] & 0xFFFF)   # [24:16]=sv_idx, [15:0]=alpha Q6.10
    wishbone_write(WB_BASE + 0x28, word)
```

Step 3 — Update gamma via PARAM_WR

```
PARAM_WR offset = 0x24
# addr=0 is gamma register; [19]=en, [18:16]=addr, [15:0]=value
word = (1 << 19) | (0 << 16) | gamma_Q6_10
wishbone_write(WB_BASE + 0x24, word)
```

Step 4 — Update SV counts if changed

Only required if the new model has different SVs per class than the previous model.

```
NUM_SV offsets: class 0 = 0x10, class 1 = 0x14, ..., class 4 = 0x20
for c in range(5):
    wishbone_write(WB_BASE + 0x10 + c*4, num_sv[c])
```

Step 5 — Fire start

```
wishbone_write(WB_BASE + 0x04, 0x0B)   # CONTROL: start=1, vbatt_ok=1, kern_ready=1
```

Constraints

- Total SV count must not exceed 500 (100 per class maximum). Models exceeding this require re-synthesis with updated `NUM_SV` parameters.
- Gamma must be representable in Q6.10 (range 0–63.999, resolution ~0.001).
- Alpha values must be representable in Q6.10 (signed, range −32 to +31.999).
- The reload can be performed between any two batches. The ASIC does not need to be held in reset during the write sequence — the alpha table and gamma register are shadow-registered and only take effect when `start` fires.
- Writing ALPHA_WR while `STATUS[done]=0` (i.e., during active inference) has undefined behavior and must be avoided.

Appendix B — Alternative Design: Hospital-Grade Batch Classifier (28nm)

This appendix specifies an alternative architecture targeting continuous bedside cardiac monitoring in a hospital environment — no power or area constraints, 100-beat batch processing, GEMM-based matrix engine.

B.1 Design Goals

Parameter	Wearable (current)	Hospital (proposed)
Use case	Ambulatory patch	Bedside monitor / ICU
Batch size	1000 beats	100 beats
Latency requirement	< 750 ms/beat	< 5 s per 100-beat batch
Power budget	< 1 mW avg	No constraint
Area budget	Caravel die (6.25 mm²)	No constraint
Process	sky130A (180nm equiv.)	TSMC 28nm HPC

B.2 Feature Set (unchanged)

The same 256-dimensional multi-scale feature vector is used:

Group	Dims	Content
Single-beat morphology	128	±64 samples, amplitude-normalized
10-beat mean template	64	Average of preceding 10 beats

Group	Dims	Content
100-beat mean template	64	Average of preceding 100 beats (replaces RR-interval history)
Total	**256**	

The 100-beat template replaces the RR-interval history for hospital use, providing longer-term morphological context available in a continuous-monitoring environment.

B.3 Architecture

Compute engine: 32×32 systolic array

A 32×32 weight-stationary systolic array processes the pairwise distance matrix as a single GEMM operation rather than sequentially iterating over SVs.

The distance computation is restructured as:

```
D[i,j] = ||X[i] − SV[j]||²
       = ||X[i]||² − 2·(X · SV^T)[i,j] + ||SV[j]||²
```

The dominant term `X · SV^T` is a **100×256 × 256×500** GEMM — 12.8M MACs, fully parallelized across the systolic array. The squared norm terms are precomputed in a single pass and broadcast.

Memory: on-chip SRAM (1 MB total)

Buffer	Size	Contents
Input matrix	51.2 KB	100 beats × 256 features × 16-bit Q6.10
SV matrix	256 KB	500 SVs × 256 features × 16-bit Q6.10
Distance matrix	200 KB	100 × 500 × 32-bit intermediate
Kernel matrix	100 KB	100 × 500 × 16-bit exp output
Alpha table	1 KB	500 × 16-bit (same as current)
Score / output	2 KB	100 × 5 × 32-bit class accumulators
Total	**~610 KB**	(rounded to 1 MB with ECC and overhead)

Clock and process

Parameter	Value
Process	TSMC 28nm HPC
Supply voltage	0.9 V
Clock	800 MHz
Peak compute	1024 MACs/cycle × 800 MHz = **819 GOPS**
On-chip SRAM bandwidth	~100 GB/s

B.4 Performance

Per 100-beat batch:

Stage	Operations	Cycles	Time
Load input + SV to SRAM	—	—	one-time, DMA
GEMM: X · SV^T (100×256×500)	12.8M MACs	~12,500	15.6 µs
Squared norms + broadcast	100K ops	~100	0.1 µs

Stage	Operations	Cycles	Time
Exp LUT (100×500 evaluations)	50K	50,000	62.5 µs
Alpha accumulation (100×500×5)	250K MACs	~250	0.3 µs
Argmax (100 × 5)	500	~1	<0.1 µs
Total per 100-beat batch		**~63,000**	**~78 µs**

Throughput: 100 beats / 78 µs = **1,280,000 inf/s (1.28 M inf/s)**

Duty cycle at 80 bpm (100 beats every 75 s): 78 µs / 75,000,000 µs = **0.000104%**

B.5 Power

Component	Active	Standby
Systolic array (32×32, 0.9V, 800 MHz)	~180 mW	~0
On-chip SRAM (1 MB active)	~80 mW	~3 mW leakage
Control logic, IO	~40 mW	~0.5 mW
Total	**~300 mW**	**~3.5 mW**

Average power at 80 bpm (100-beat batches):

```
P_avg = P_active × duty_cycle + P_leakage
      = 300 mW × 0.000104% + 3.5 mW
      ≈ 0.0003 mW + 3.5 mW
      = ~3.5 mW
```

Average power is dominated entirely by SRAM leakage. The compute contribution is negligible — the chip spends 99.9999% of the time idle.

B.6 Die Area

Block	Area (28nm est.)
1 MB on-chip SRAM	~0.50 mm²
32×32 systolic array	~0.30 mm²
Horner LUT + exp pipeline	~0.05 mm²
Control FSM, IO, misc	~0.10 mm²
Total estimated	**~1.0 mm²**

The hospital design is **6× smaller die area** than the current sky130A core (6.25 mm²) despite being orders of magnitude more capable — a direct consequence of the 28nm process density advantage.

B.7 Roofline Comparison

The fundamental shift from the wearable to the hospital design is the operational intensity: the GEMM formulation reuses the SV matrix across all 100 input beats, dramatically increasing arithmetic intensity.

Operational intensity:

Design	Total ops	Memory traffic	Ops/byte
Wearable (sequential, per beat)	512K ops	256 KB (SV re-read each beat)	**2.0**
Hospital (GEMM, per 100-beat batch)	25.6M ops	307 KB (SV loaded once)	~**83**

The wearable design sits deep in the **memory-bound** region of the roofline — 98.9% of time waiting for SRAM data. The hospital design operates well above the ridge point (~16 ops/byte at 100 GB/s / 1.6 TOPS) and is firmly **compute-bound**.

B.8 Comparison Table

Throughput and latency:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Throughput	309 inf/s	**1,280,000 inf/s**	4,200 inf/s	95,000 inf/s
Latency / 100 beats	323 ms	**0.078 ms**	23.8 ms	1.05 ms
Speedup vs sklearn	0.07×	**305×**	1×	22.6×

Power, area and roofline:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Active power	66 mW	300 mW	15,000 mW	80,000 mW
Avg power (80 bpm)	**0.284 mW**	3.5 mW	~15,000 mW	~80,000 mW
Die area	6.25 mm^2	~1.0 mm^2	N/A	N/A
Process	sky130A (180nm)	TSMC 28nm	Intel 10nm	Intel 10nm
Ops/byte	2.0	**83**	~2.0	~2.0
Roofline regime	Memory-bound	**Compute-bound**	Memory-bound	Memory-bound

Key observations:

- The hospital ASIC is **305× faster than sklearn** and **13.5× faster than 8-core Numba** on a 100-beat batch
- This is achieved by converting a memory-bound sequential problem into a compute-bound GEMM
- The SRAM leakage floor (3.5 mW) makes the hospital design unsuitable for wearable use — 12× worse average power than the current design despite being 4,000× faster
- Both designs achieve 97.67% accuracy — the architecture difference is entirely in throughput and power, not classification quality